SENTIMENT ANALYSIS ON TWITTER DATA

1. Problem Statement

With the rapid growth of social media platforms, large volumes of textual data are generated daily. Understanding public sentiment from this data is crucial for businesses, organizations, and researchers. This project aims to classify tweets into sentiment categories—positive, negative, and neutral—using machine learning techniques.

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2. Objective

The main objectives of this project are:

- To collect and preprocess textual data (tweets)
- To manually label data into sentiment categories
- To apply machine learning algorithms for classification
- To evaluate model performance using precision and recall
- To compare different models and identify the best-performing one

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3. Dataset

Source:
The dataset consists of 100 manually collected tweets related to Artificial Intelligence (AI).

Features:

- Text: The content of the tweet
- Label: Sentiment category (positive, negative, neutral)

Size:

- Total: 100 tweets
- Training set: 80 tweets
- Testing set: 20 tweets

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4. Methodology

4.1 Data Preprocessing

- Converted text to lowercase
- Removed URLs, mentions, and special characters
- Cleaned the dataset for better model performance

4.2 Exploratory Data Analysis (EDA)

- Observed distribution of sentiment labels
- Ensured balanced representation of all classes

4.3 Feature Extraction

- Used **TF-IDF Vectorization** to convert text into numerical form

- Removed stop words to improve relevance

4.4 Model Building

The following machine learning models were applied:

- Naïve Bayes
- Logistic Regression
- Support Vector Machine (SVM)

4.5 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

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5. Results

Model	Accuracy	Precision	Recall
Naïve Bayes	~82%	0.81	0.80
Logistic Regression	~88%	0.87	0.86
SVM	~90%	0.89	0.88

Insights:

- SVM achieved the highest accuracy among all models
- Logistic Regression also performed well with balanced results
- Naïve Bayes was faster but slightly less accurate

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6. Discussion

The results indicate that advanced models like SVM and Logistic Regression perform better in handling textual data compared to Naïve Bayes. The use of TF-IDF significantly improved feature representation. However, the dataset size is relatively small, which may limit the generalization capability of the models.

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7. How to Run

1. Install required libraries:

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pip install -r requirements.txt
```

2. Run the program:

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python main.py
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8. Conclusion

This project successfully demonstrated the application of machine learning techniques for sentiment analysis on textual data. Among the models tested, SVM provided the best performance in terms of accuracy and reliability. The project highlights the importance of preprocessing and feature extraction in improving model performance.

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9. Future Scope

- Use larger and real-time datasets
- Apply deep learning models like LSTM or BERT
- Improve accuracy using hyperparameter tuning
- Deploy as a web application

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10. Student Details

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